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Social Simulation of COVID-19 with AI in Japan

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Projection of COVID-19 Positive Cases Considering New Viral Variants and Vaccination Effectiveness Models: Deep Learning Approach

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1 Introduction

1.1 Background

Since its emergence in late 2019, SARS-CoV-2 has spread rapidly across the globe, resulting in widespread hospitalizations and fatalities [1]. The transmission patterns of the virus have been influenced by the evolution of variants and sublineages with differing infectivity, as well as public behaviors shaped by regional policies, cultural practices, and societal attitudes [2, 3]. Countries in the Asia–Pacific region, such as Hong Kong, Japan, New Zealand, Singapore, and South Korea, generally implemented stricter behavioral policies [4] than many European and American nations [5, 6].

The progression and decline of COVID-19 waves varied significantly across countries [7]. During the first wave, China's strict lockdown measures led to a rapid reduction in positive cases [8, 9]. Similarly, Japan experienced a swift rise and fall in cases during its first wave, which was attributed to the implementation of a state of emergency [10]. In contrast, the decline phases of waves in European countries, such as the United Kingdom and Germany, were generally more prolonged than those in Asian countries [7].

The spread of infectious diseases, including COVID-19, is heavily influenced by immunity, particularly adaptive immunity. In this context, immunity is defined as the

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level of antibody-mediated protection against symptomatic infection, encompassing both vaccine-induced immunity and naturally acquired immunity from previous infections. Adaptive immunity, however, wanes over time following infection or vaccination [11]. Tang et al. [12] provide a detailed summary of population-level immunity in various countries as of early 2021. Rather than achieving herd immunity, transient suppression of transmission has been suggested as a more feasible outcome [13], emphasizing the importance of localized approaches and targeted interventions over broad, nationwide strategies. Thus, an analytical or computational approach is needed, considering these aspects, to forecast new daily positive and death cases as well as to support the interpretation of the results.

1.2 Vaccination and COVID-19 Situation in Japan

Vaccination efforts began in December 2020 in Israel, leading to a sharp decline in positive cases by January 2021 [14]. mRNA vaccines demonstrated high efficacy, providing over 90% protection against symptomatic infection by the Delta variant two weeks after full vaccination [15–18]. However, their effectiveness waned against the Omicron variant and its subvariants due to the immune evasion properties of these strains [16, 19, 20]. Additionally, several countries, including Taiwan, Australia, South Korea, and Japan [1, 21], experienced substantial surges in new cases starting in late 2021 and early 2022. These surges were largely driven by viral variants with greater infectivity and shorter generation times compared to earlier variants [22, 23]. These findings highlight the critical need to monitor spatiotemporal variations in population-level (adaptive) immunity to more accurately estimate viral transmission dynamics.

In Japan, the SARS-CoV-2 pandemic began in early 2020, with waves categorized as the first through seventh based on the cycles of reported daily positive cases (DPCs) [24]. The dominant variants during these waves were identified as follows: the Wuhan strain during the first to third waves, the Alpha variant in the fourth wave, the Delta variant in the fifth wave, Omicron BA.1 and BA.2 in the sixth wave, and Omicron BA.5 in the seventh wave. Before the seventh wave, the proportion of reported positive cases relative to the total population was limited to 11% (estimated from [24]), and a nationwide vaccination campaign had been launched in March 2021. These factors, along with the uniformity of Japan's health insurance and healthcare system, played a critical role in preventing the collapse of medical resources during the pandemic. By the fourth and fifth waves, the cumulative number of positive cases in Japan had reached approximately 785,000 and 1,750,000, representing 0.63% and 1.4% of the total population, respectively.

A serologic survey conducted in Tokyo estimated that the actual number of infections was 3.9 times (95% CI: 3.0–7.0) higher than the reported positive cases [25]. Despite accounting for asymptomatic infections, the estimated proportion of infected individuals remained well below the herd immunity threshold, which ranged from 50% to 80% [11, 26, 27]. Thus, the decline observed up to the

fourth wave, when vaccination rates were still low, was primarily attributed to a reduction in the effective reproductive number. This reduction was largely influenced by behavioral changes induced by state-of-emergency measures, such as those implemented in Tokyo from April 25 to June 20, 2021, during the fourth wave [3].

The administration of the third vaccination (booster dose) began in Japan in January 2022, coinciding with the emergence of the Omicron BA.1 variant. Unlike earlier waves, the sixth and seventh waves affected a substantial portion of the population, with the Omicron BA.1 variant and its BA.5 subvariant emerging as the dominant strains, respectively. Assessing transient suppression across three metropolitan prefectures in Japan and incorporating the estimated parameters into transmission models provide valuable insights for developing future policies and interventions in other regions [27].

1.3 Purpose

This chapter aims to evaluate population-level immunity during the fifth to seventh waves of COVID-19 across all three metropolitan prefectures [28], with a particular focus on vaccination status and the waning effects of immunity [18]. The analysis is designed to be systematic and highly quantitative, ensuring its suitability as input for machine learning models. We performed detailed analyzes in three major metropolitan prefectures—Tokyo, Osaka, and Aichi—to gain insights into the propagation and decline patterns of COVID-19 waves. To validate the numerical estimates of the temporal evolution of population-level immunity, we replicated the DPCs in these prefectures using machine learning techniques [29].

2 Network Architecture Based on Machine Learning

Numerous prediction models, including SIR/SEIR frameworks [30] and deep neural networks, have been developed to forecast new positive cases during epidemics. While these models provide valuable insights, their predictive accuracy is highly dependent on the quality of the data used for calibration. Furthermore, when contact rates are suppressed under a state of emergency, modeling and parameter extraction become more complex, as highlighted by studies employing multi-agent simulations [31, 32]. Machine learning models enable more flexible data modeling allowing various measurements to be considered as potential factors to address future prediction problems. During the training phase, deep neural networks can process multiple data inputs and adjust network parameters to represent data dependencies and their potential contributions to known outputs. The trained network can then be used to predict future trends related to the pandemic through data analysis and

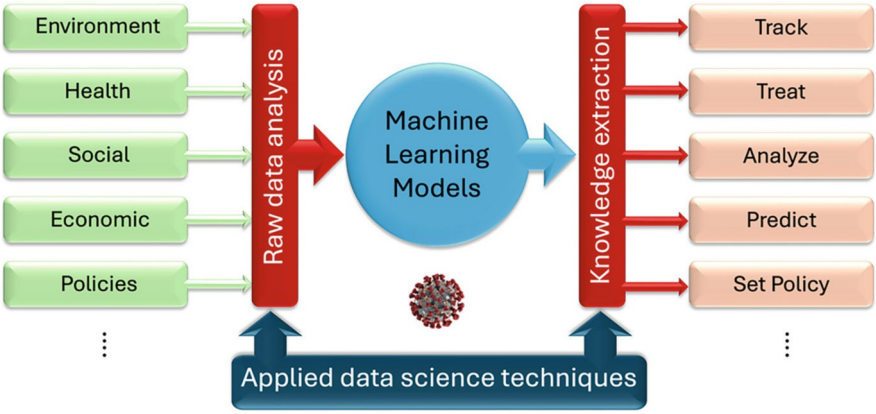

Fig. 1 Machine learning enable extracting knowledge from different data sources and analyze data dependances with several future predictions associated with pandemics

knowledge extraction processes. Figure 1 illustrates an overview of a forecasting model using machine learning approaches.

Deep neural networks are widely utilized for forecasting DPCs and death counts. A significant advantage of deep neural networks is their capacity to identify and model correlations between various factors and the incidence of infections and deaths. In [29, 33], we introduced a multipath neural network architecture that integrates long short-term memory (LSTM) modules, which are primarily used for time-series forecasting, with fully connected layers designed to capture correlation features. This architecture operates in two distinct phases and was developed based on insights gained from an ablation study.

The training process follows the procedure described in [34]. The input values, as detailed above, spanning a certain period (14 days), are used along with the target output values for the successive 14-day period to train the model. The training is conducted by minimizing the cross-entropy loss function using the Adam algorithm over 1000 epochs, with the training rate automatically estimated.

Deep learning models are commonly presented as a black box, where the contribution of different factors is unknown. To mitigate this limitation, we mathematically derived population-level immunity (Sect. 3.2). In [35], the optimal combination of input values was investigated, including mobility, meteorological data, parameters related to social behavior, day labels, population vaccine effectiveness (VE), and variant infectivity. The selection of the inputs was based on their correlation with morbidity and several potential factors.

The input data were structured in a 2D array, where the x-axis represented the time-series data and the y-axis indicated the values of different input variables. The target output was defined as the DPC associated with each date. The following factors were selected as input values based on the mean absolute percentage error (MAPE) by comparing the reported and estimated DPC: mobility at transit stations,

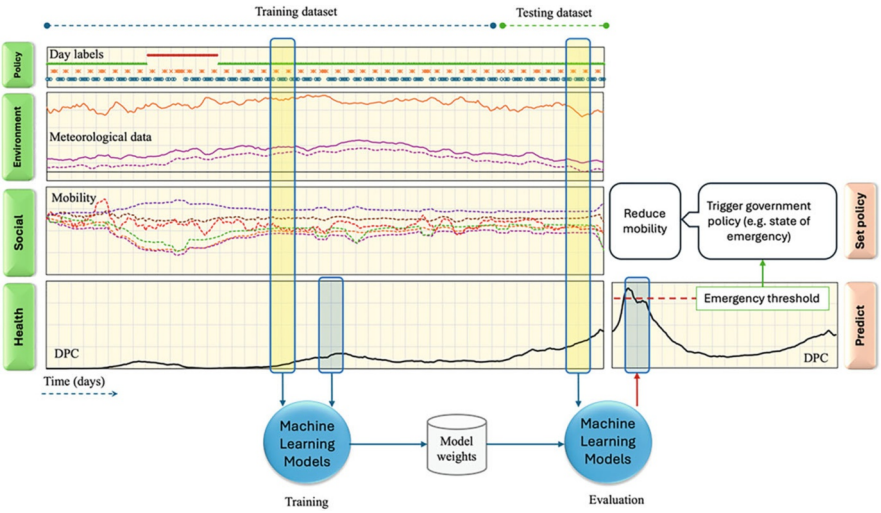

Fig. 2 Data measurements including different data types are used to train deep neural network to estimate future DPC. The estimated DPC with specific threshold can be used to alert policy maker to set a new policy (for example, to restrict public mobility)

the number of tweets containing risk-related keywords on Twitter (renamed as X), and population VE, which was calculated from individual VE and vaccination rates [23]. A demonstration of the DPC prediction using a deep neural network is shown in Fig. 2.

3 Data

3.1 Number of Reported Positive Cases per Day

The DPC data for each prefecture were sourced from open data provided by the Ministry of Health, Labor and Welfare [24]. The number of DPCs is affected by factors such as the incubation period and the day of the week, with typically fewer tests conducted on weekends than on weekdays. To minimize the impact of these variations, we applied a 7-day moving average (± 3 days from the corresponding day) to smooth the data. Using the smoothed 7-day moving average of the DPCs, we identified the peak date as the day with the highest value for each wave. We assumed that each wave followed a Gaussian distribution and numerically fitted the data using the least squares method to analyze the time course of the 7-day average DPCs. Based on the fitted curve, we defined the start of a wave as the date when the curve's peak value first exceeded 10%. In some prefectures, the decay phase of one wave overlapped with the propagation phase of the subsequent wave.

3.2 Vaccination Effectiveness

3.2.1 Effectiveness of Individual Vaccination

The daily number of individuals vaccinated (from the first to fourth doses) was obtained from the Vaccination Record System of the Digital Agency, Japan [29]. Vaccination in Japan began in March 2021 for medical workers, followed by older adults in April, and nonmedical workers in June, with vaccinations administered nearly uniformly nationwide. The interval between the first and second doses was maintained at 3–4 weeks. The administration of third and fourth doses began on December 1, 2021, and May 25, 2022, respectively.

Individual vaccination effectiveness (VE) was defined as VE = 1 − relative risk in the real world without controlling for conditions, as follows:

$$e_t(i) = \begin{cases} a_t \cdot i/K \ \ (i \leq K), \\ a_t - s\,(i - K) \ \ (i > K), \end{cases} \tag{1}$$

$$e_{inf}(i) = a_{inf} - s \cdot i, \tag{2}$$

where e_t (i) represents the effectiveness of vaccination of an individual on day i after the t-th vaccination ($t = 1, 2, 3, 4$). This linear equation [18] is derived from the logistic relationship between neutralization level and protective efficacy [30]. This assumption is consistent with findings from a cohort study of VE in Japan [36].

The effectiveness of vaccination for an individual reaches its peak a_t, K days after vaccination ($K = 14$ for the second dose, $K = 7$ for the third and fourth doses). The parameters a_t for each variant were derived by comparing the prevalence of COVID-19 among unvaccinated individuals and those without immunity to prevent infection in our previous study [31]. The waning immunity s from vaccination was approximated linearly. Similarly, the waning effect of VE against symptomatic infection s was also approximated linearly. The function e_{inf} (i) represents the natural immunity of an individual on day i after infection, which wanes as day i progresses from the peak a_{inf}. The parameters a_t for each variant, a_{inf}, and s were used as described in [37].

The reported number of positive cases is influenced by latency, which is a combination of the incubation period and the delay in reporting from hospitals or clinics to the prefecture. When analyzing the correlation between mobility and the effective reproduction number in relation to positive cases, the latency (the lag between reported social behavior and the number of DPCs) was estimated at 10 days for the Delta variant [3] and 8 days for the Omicron variant and its subvariants. To evaluate population-level immunity appropriately, the immunity levels from 8 to 10 days prior to the peak date of DPC was used in the analysis.

3.2.2 Estimation of Population-Level Immunity

Population-level immunity is defined as the combination of VE at the population level and the estimated proportion of infected cases within the entire population. In a serological survey conducted in Tokyo during the third wave (end of March 2021), the number of infected cases was estimated to be 3.9 times higher (95% CI: 3.0–7.0) than the number of positive cases [25].

As part of surveillance efforts in Tokyo beginning on April 1, 2021, polymerase chain reaction (PCR) tests were conducted on asymptomatic individuals in various locations, including downtown areas, restaurants, airports, and major train stations, to estimate the population with asymptomatic infections [38].

Population-level immunity E is defined as the combined VE at the population level and the estimated ratio of infected cases within the entire population:

$$\begin{aligned} E(d) = & \sum_{t=1}^{T} \sum_{i=0}^{d} \sum_{s} \left(N_t\,(d-i) \cdot e_t(i) v_s(d)\right) / P \\ & + \sum_{i=0}^{d} \sum_{s} \left((1+f) \cdot \mathrm{DPC}\,(d-i)\, v_s(d) \cdot e_{inf}(i)\right) / P \end{aligned} \tag{3}$$

where the first term on the right-hand side represents the VE at the population level. The second term accounts for natural immunity resulting from infections, assuming that natural immunity is effective only against the same variant. In Eq. (3), d denotes the day index, N_t refers to the number of individuals newly administered a vaccination dose ($t = 1$–4), P represents the population of each prefecture, and v_s represents the rate of each variant of SARS-CoV-2 on the d day. This equation is derived from Eqs. (1) and (2) considering the prefecture-specific populations and the waning effect of vaccination. The DPC represents the number of cases reported on a given day, and parameter f denotes the rate of asymptomatic individuals, assumed to be four to eight times greater than that of symptomatic cases.

The SARS-CoV-2 sequences by variants are shown in Fig. 3a. Figure 3b depicts the time course of the ratio of estimated infected cases to confirmed positive cases based on PCR tests. From the figure, the ratio varies depending on the time window which is consistent across other variants [32]. The ratio tends to become relatively unstable when the number of DPCs is small. It should be noted that different operators conducted surveillance activities before and after April 2022. Figure 3c illustrates the vaccination rate in Japan over time.

Empirically, it is assumed that the number of infected cases is calculated as the number of reported cases multiplied by a factor of 6 (3–5 for the Omicron variant and 7–11 for the BA.5 variant). Due to the lack of similar surveys in other regions, this assumption has been applied uniformly across all prefectures. It is worth noting that this ratio is close to 3.0–7.0 times (95% CI) obtained in earlier waves of the

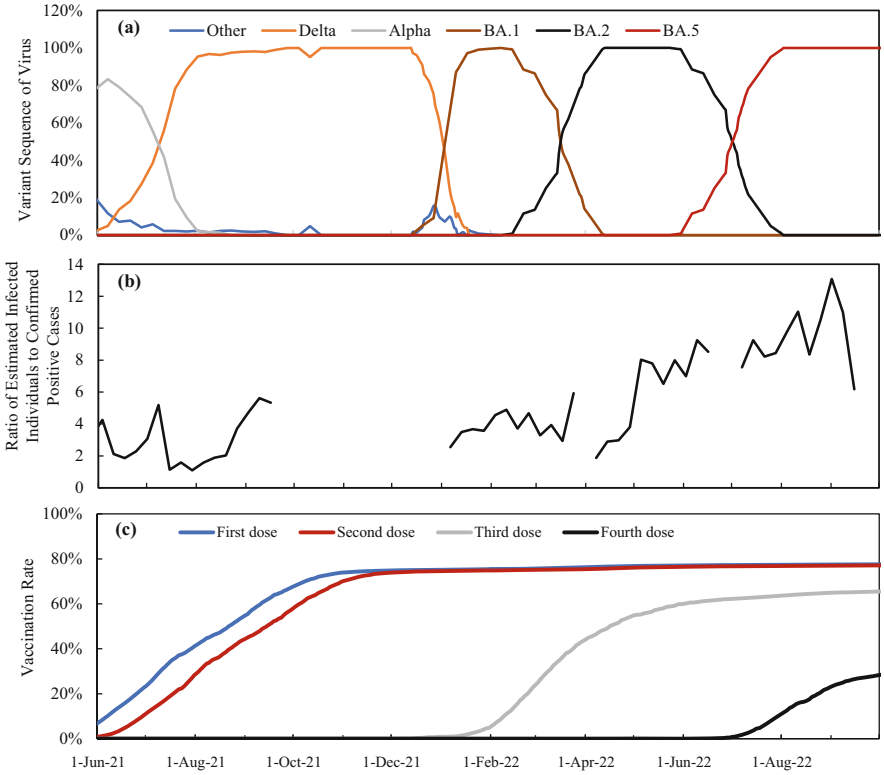

Fig. 3 Time course of (**a**) dominant viral variants, (**b**) ratio of infected individuals without symptoms to empirically derived from the report in Tokyo, and (**c**) vaccination rate in Tokyo. In (**b**), the data for periods with less than 500 DPCs are excluded due to considerable uncertainty. (Kodera et al., *Vaccines*, **11**, 1457, 2023)

cohort study [24], although this ratio may vary depending on age groups and viral variants.

3.2.3 Waning Effect of Vaccination among the Japanese Population

Figure 4 shows the waning effect of vaccination among the Japanese population aged 65 years and older and those under 65 years. As shown in Fig. 4a, waning immunity was approximately linear for individuals in both age groups. Due to the shorter elapsed time for individuals younger than 65 years, the difference in waning immunity between the two groups was unclear, although an IEV of approximately 95% was estimated immediately after the second dose. The coefficient of determination was 0.9984 ($p < 0.0001$) for individuals under 65 and 0.9927 ($p < 0.0001$) for those over 65 years of age.

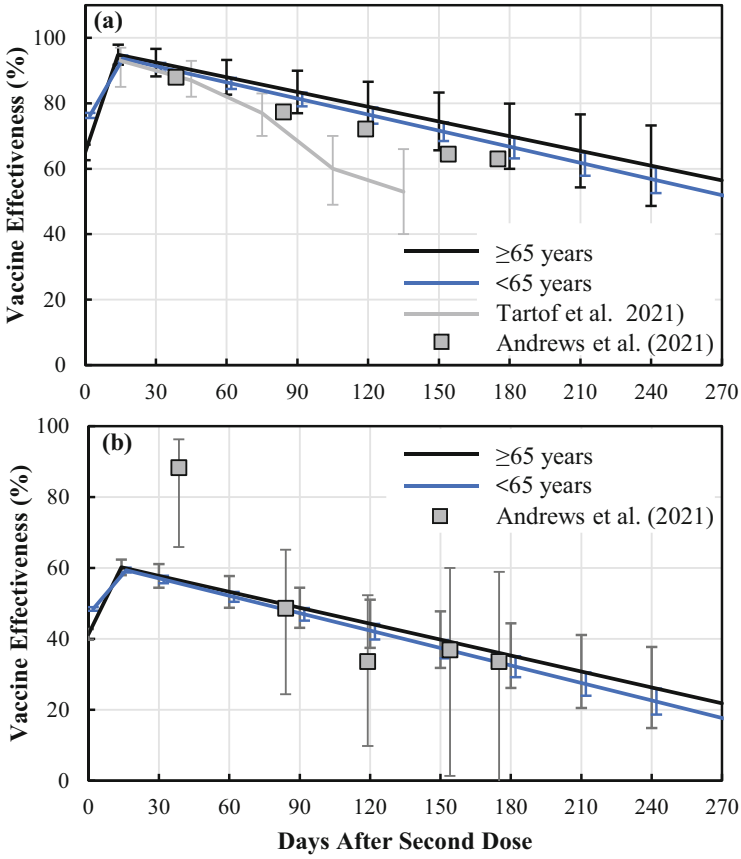

Fig. 4 Waning effect of vaccination in Japan, for individuals over and under 65 years of age, with respect to protection against symptomatic infection by (**a**) Delta and (**b**) Omicron variants derived for Japanese population. Bars represent the standard deviation. (Kodera et al., *Vaccines*, **10**, 430, 2022)

Based on the ratio of positive cases between non-vaccinated and fully vaccinated individuals for the Delta and Omicron variants, the waning effect for the Omicron variant was estimated as 64.5% (95% CI: 50%–68%) of that for the Delta variant. In the following discussion, the IEV for protection against infection with Omicron is assumed as depicted in Fig. 4b. The parameters used in Eq. (1) are presented in Table 1.

Table 1 Parameters for the individual effectiveness of vaccination used in Eq. (1) for delta and omicron variants (Kodera et al., *Vaccines*, **10**, 430, 2022)

	Delta		Omicron	
	Mean	95%CI	Mean	95%CI
	All age category			
a_1	74.9	(70.8–78.9)	48.3	(35.6–53.7)
a_2	96.2	(95.6–96.9)	62.1	(48.0–66.0)
s	0.149	(0.132–0.165)	0.149	(0.132–0.165)
	Under 65 years of age			
a_1	76.3	(73.9–78.7)	49.2	(37.1–53.6)
a_2	93.8	(93.1–94.6)	60.5	(46.7–64.4)
s	0.164	(0.152–0.177)	0.164	(0.152–0.177)
	Over 65 years of age			
a_1	65.0	(70.8–78.9)	41.9	(22.9–57.5)
a_2	95.0	(95.6–96.9)	61.3	(46.5–66.3)
s	0.150	(0.111–0.189)	0.150	(0.111–0.189)

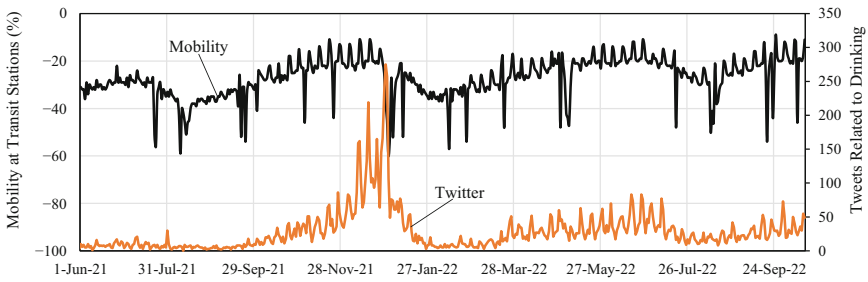

Fig. 5 Time series of mobility at transit stations and the volume of drinking-related tweets on Twitter (X) in Tokyo. (Kodera et al., *Vaccines*, **11**, 633, 2023)

3.3 Mobility at City Centers

3.3.1 Data Source

Mobility data were collected from Google Mobility [24]. Mobility is defined as the percentage difference in population volume at transit stations compared to a baseline. The baseline represents the median value for that day of the week over the 5 weeks from January 3 to February 6, 2020 (before the COVID-19 pandemic). Twitter data were obtained from NTT Data, Inc.; processed by the Toyoda Lab., University of Tokyo; and shared through the Cabinet Secretariat COVID-19 AI & Simulation Project. Keywords from tweets posted during the day, the previous day, or planned for the next day were extracted when determining the number of tweeted keywords. Time-series data for mobility and the number of tweets are summarized in Fig. 5.

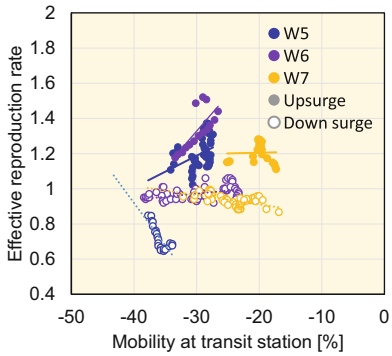

Fig. 6 Correlation between the mobility at the transit station and the effective reproduction rate in Tokyo. Mobility was averaged over 8 days with a latency of 6 days

3.3.2 Correlation with Effective Reproduction Number

Figure 6 shows the correlation between the ERN and the average mobility at transit stations during the upsurge and downsurge of DPC, with the duration and latency set to 8 and 6 days, respectively. A positive correlation was observed between the ERN and the time-window-averaged mobility during the upsurge, while a negative correlation was observed during the downsurge of positive cases in the fifth and sixth waves.

The slopes of the regression lines are comparable between different variants. The ERN for the Omicron variant was approximately 20% higher than that of the Delta variant in Tokyo. However, a direct comparison remains difficult between the variants due to the increased rate of vaccination campaigns during and after the spread of the Delta variant.

3.4 Data Associated with Social Behavior

We considered three metrics to assess the effect of social behavior on the individual VE: (1) mobility at the transit stations, (2) nighttime population who stayed in the downtown area including restaurants and bars [39], and (3) Twitter keywords (social gathering for drinking and BBQ). These are the human behavior metrics made available by the Cabinet Secretariat COVID-19 AI Simulation Project (https://www.covid19-ai.jp/en-us/). The rationale for these metrics is based on a study [40] that found that most infections can be traced back to specific locations, particularly restaurants. If these metrics can explain the daily variation in VE, they would be excellent candidates for long-term monitoring.

The nighttime population in areas near restaurants and bars was obtained from NTT DOCOMO, Inc., and made accessible through the Cabinet Secretariat COVID-19 AI Simulation Project, Japan. Twitter data was utilized as it may reflect social

activities where close contact occurs. The downtown population was included in the analysis, as several domestic reports have suggested that the main infection clusters likely resulted from close contacts in these areas. The time-series data for these parameters are demonstrated in Fig. 7.

Twitter (now X) data were obtained from NTT Data, Inc., processed by the Toyoda Lab at the University of Tokyo, and shared through the Cabinet Secretariat COVID-19 AI Simulation Project. Keywords from tweets made on the same day, the previous day, or planned for the following day were extracted to determine the number of relevant tweets. The recorded data clearly indicates timeframes when these events were more popular, although it remains challenging to confirm whether these gathering events were actually held. For the corresponding tweets, information about the prefecture was extracted from the user's registered address.

A 7-day average value was used to account for the weekly effect when analyzing the correlation with VE. Time lags of 8 and 5 days for the three behavioral metrics were also considered during the New Year holidays and Golden Week, respectively. These lags correspond to the incubation period and reporting delays for the Delta and Omicron variants [3].

To assess the effect of human behavior on population VE, statistical analysis was conducted using Python 3.9.7. A linear regression model was applied to fit the drop in estimated population VE as the dependent variable against each independent variable: the number of tweets, nighttime population, and mobility at transit stations. The Spearman's rank correlation test was performed as well. The threshold for statistical significance was set at $p < 0.05$.

3.5 Data Associated with Weather

Other factors that may influence viral transmission include meteorological data [41, 42], which were extensively discussed in the early stages of the pandemic [43–45]. However, correlations between viral transmission and human behavior were more prominently reported during the later stages of the pandemic, particularly with the advent of vaccination campaigns and the emergence of viral variants [46, 47]. According to [35], the contribution of meteorological factors is marginal for mid- and long-term projections compared to other factors. The association between meteorological factors and viral infection may also be closely tied to social behavior [44], as infections in metropolitan areas often occur indoors. As a result, meteorological factors were not considered in this analysis.

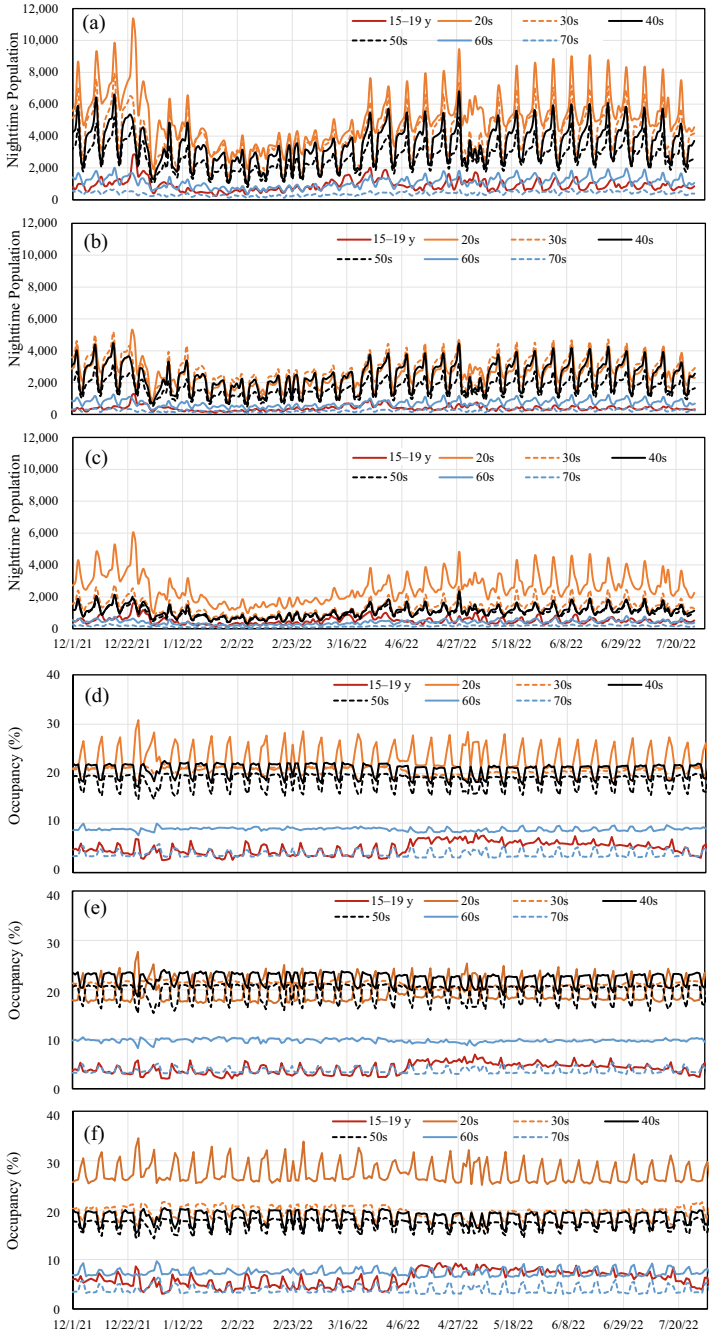

Fig. 7 (**a**–**c**) Time course of the nighttime population by age category and (**d**–**f**) the occupancy of age category in the mobility at the Tokyo stations. The gender categories are (**a**, **d**) all, (**b**, **e**) male, and (**c**, **f**) female. (Kodera et al., *Vacciness*, **10**, 1820, 2022)

4 Results

4.1 Daily Reported Positive Cases and Estimated Population-Level Immunity

Figure 8 shows the DPCs and population-level immunity estimates for Tokyo, Osaka, and Aichi. Considering that asymptomatic infections are estimated to be four to eight times greater than reported DPCs, the numbers reflect the uncertainty in immunity efficacy (Fig. 2b). Before the sixth wave, the impact of the infected population on population-level immunity was minimal, but it became relatively significant thereafter. This trend was similarly observed in Osaka and Aichi.

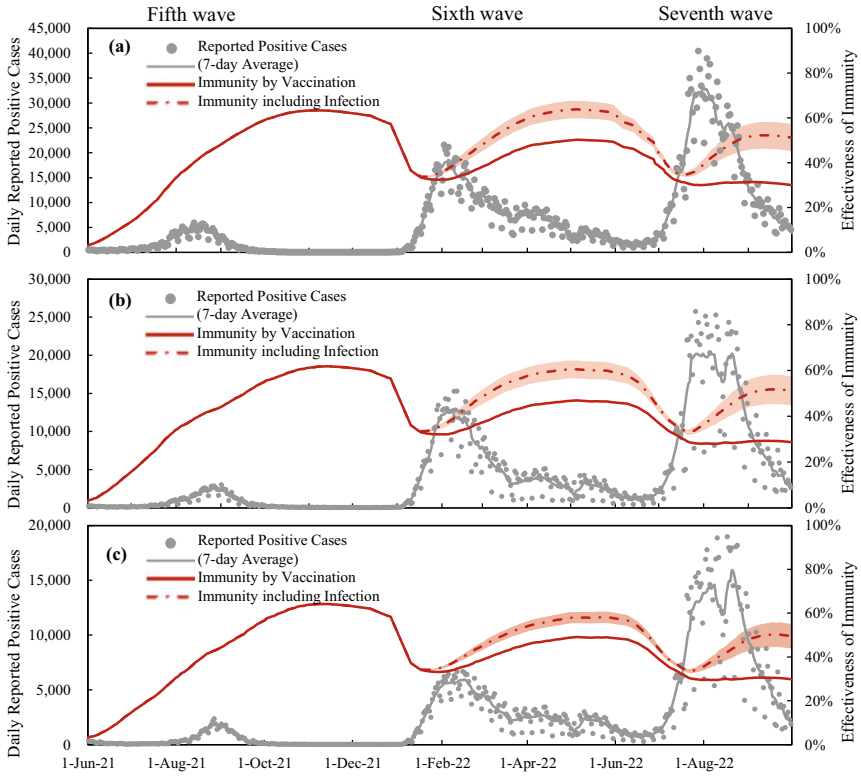

Fig. 8 Daily reported positive cases and estimated population-level immunity in (**a**) Tokyo, (**b**) Osaka, and (**c**) Aichi. The dashed line indicates population-level immunity assuming that the number of asymptomatically infected individuals is six times the number of positive cases reported daily. The colored area represents the variation in the immunity effectiveness due to the variation in the number of asymptomatically infected individuals (4–8 times). (Kodera et al., *Vaccines*, **11**, 1457, 2023)

As a result, comparable levels of population-level immunity were required in the three prefectures during the fifth to seventh waves. For all waves, the required level of population-level immunity for suppression was approximately 40%. The sixth and seventh waves showed a gradual increase in immunity, despite immunity for the entire population remaining relatively similar. Notably, for the fifth wave, the immunity of the population aged 10–64 years required for peak formation was less than that of the entire population. This is attributable to the early vaccination strategy, which prioritized the older population (see Sect. 1.2). The percentage of fully vaccinated individuals aged 10–64 years was approximately 20% at the peak of the fifth wave, whereas for those aged ≥65 years, it exceeded 80% [48].

4.2 Verification and Interpolation with Machine Learning

Using the estimated effectiveness of immunity, we replicated daily new cases using machine learning. In this replication, future mobility and the number of tweets were assumed to be known variables. Figure 9 illustrates 1-month forecast examples in Tokyo, Osaka, and Aichi, comprising 15 time frames. The mean absolute percentage error for the forecast after 1 month was 23.7% in Tokyo, 28.1% in Osaka, and 27.3% in Aichi for days with more than 500 DPCs.

From these results, machine learning proved useful for assessing the impact of tweets, mobility, and VE on new DPCs. As a hypothetical scenario, we assumed that no new vaccines were administered after July 11, 2021. As shown in Fig. 9, a further increase in DPCs was projected in the absence of vaccination.

4.3 Model Application of Hypothetical Scenarios

From these results, machine learning proved to be a valuable tool for assessing the impact of the number of tweets, mobility, and VE on new DPCs. We considered two hypothetical scenarios.

In the first scenario, the vaccination campaign begins 1 month earlier than it actually did. This hypothesis accounts for the vaccine shortage immediately following the production of mRNA vaccines and the emergence of the Delta variant prior to the Olympic Games. If the spread of the Delta variant had been suppressed, the situation during the Olympic Games might have been different. To explore this, we simulated the fifth wave, spanning June 1, 2021, to November 30, 2021.

The second scenario assumes that prevention measures to curb the spread of the disease were implemented before the emergence of the Omicron variant. In Japan, the Omicron variant appeared and spread rapidly just after the New Year season. An open question remains: if prevention measures had been enforced before the variant's emergence, could its spread have been effectively suppressed? To simulate this, we predicted the number of new positive cases by multiplying the number of

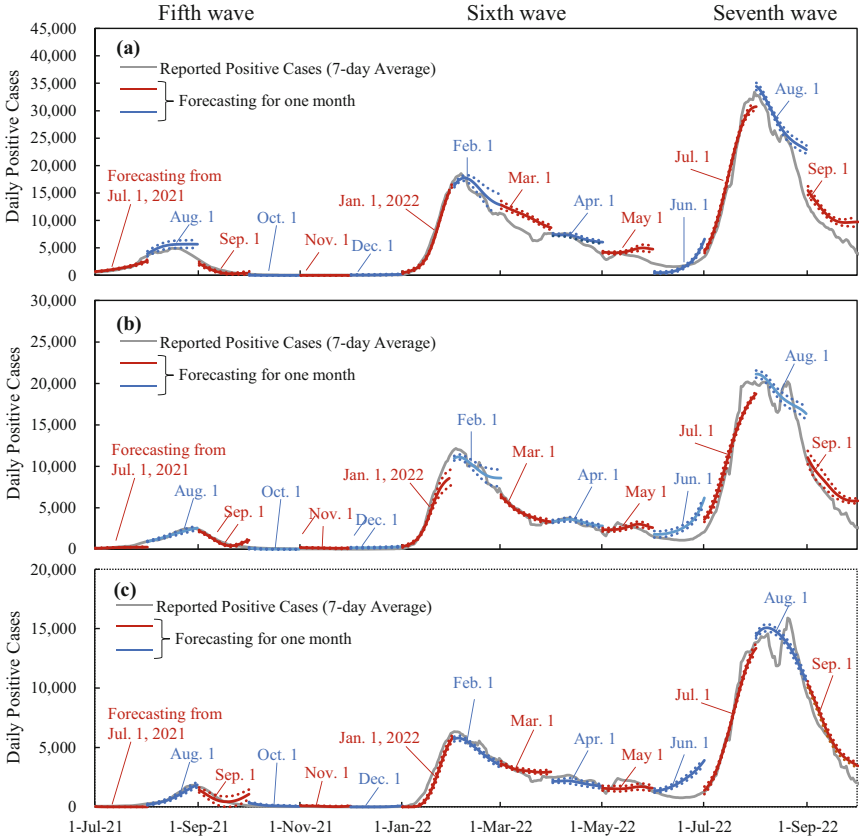

Fig. 9 Performance of the prediction of daily positive cases in (**a**) Tokyo, (**b**) Osaka, and (**c**) Aichi. In the forecasting, Twitter and mobility were assumed to be known from the reported values, and the estimated population-level immunity in Fig. 2 was used. The dotted lines represent 95% CI (Kodera et al., *Vacciness*, **11**, 1457, 2023)

tweets related to drinking by factors of 0.5 and 0.7, representing scenarios with fewer social gatherings.

As shown in Fig. 10 (a), the total number of positive cases during the fifth wave (221,124 cases) is estimated to decrease to 122,985 cases when the vaccination campaign starts 1 month earlier, while the peak number of new DPCs would be suppressed by 67%. From Fig. 10 (b), the total number of positive cases during the sixth wave (1,158,167 cases) is estimated to decrease to 633,888 cases, with the peak number of DPCs reduced by up to 46.3% through behavioral restrictions.

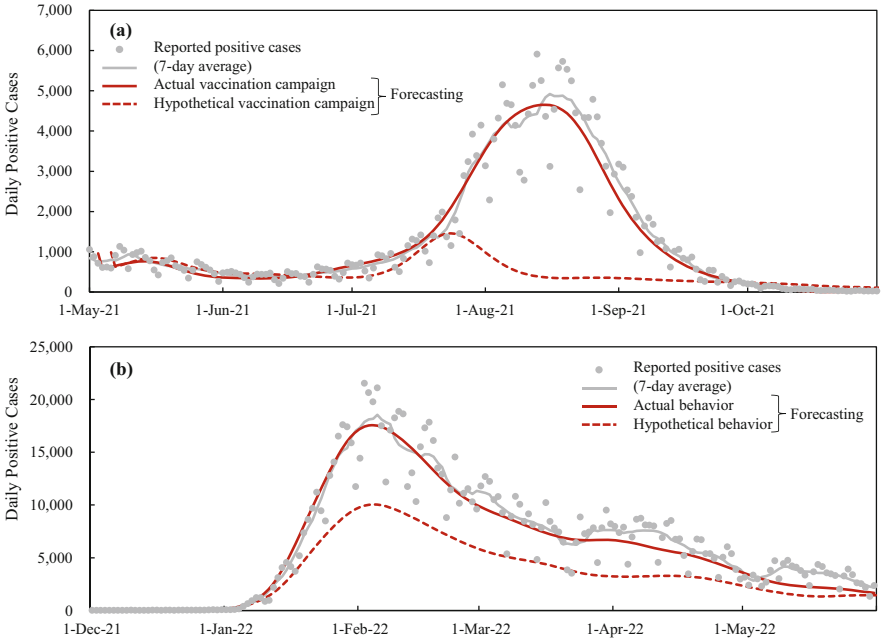

Fig. 10 Forecasting daily positive cases in Tokyo based on actual vaccination administration and the drinking-related tweets (solid line) and a hypothetical scenario (dashed line) (**a**) fifth wave and (**b**) sixth wave. The hypothetical scenarios considered are: (**a**) the vaccination campaign commencing 1 month earlier, and (**b**) behavioral restrictions resulting in a 50% reduction in drinking-related tweet volume

5 Discussion

This chapter reviewed the modeling of COVID-19 spread and decay using machine learning. Specifically, it focuses on population-level immunity required for transient suppression during Japan's fifth, sixth, and seventh waves, when the Delta variant, Omicron variant, and BA.5 subvariant were predominant, respectively.

We developed an analytic approach to estimate protection against symptomatic infections, which is essential for assessing population-level VE (Fig. 4). Additionally, we reviewed parameters that potentially influence viral transmission, including social behavior metrics such as mobility and sentiment analysis. The effectiveness of some parameters was demonstrated using statistical analysis.

From Fig. 8, in the fifth wave, the population-level immunity required to suppress the spread of infection was approximately 40% for the entire population and the population aged 10–64 years, respectively. Most cases were found for the population aged 10–64 years; therefore, an immunity level of 20% would be close to the immunity required in reality under the condition of state of emergency [28]. Unlike the later wave, mobility for the population aged 10–64 years was smaller in the fifth

wave than in the later waves due to the declaration of the state of emergency. The reduction in mobility in downtown areas was approximately 40%, 25%, and 20% lower than the levels in Tokyo, Osaka, and Aichi, respectively (see Fig. 7). When social activity is low, the transmission of SARS-CoV-2 may also be suppressed.

To validate the aforementioned heuristic method, we used machine learning to estimate the DPC. As depicted in Fig. 9, the prediction based on the assumption that future behavioral and vaccination parameters are known was in good agreement with the cases reported by the three prefectures. This comparison suggested that the heuristic method based on the numerical derivation of immunity was appropriate. Note that our vaccination parameters and the prediction model were verified to be acceptable using machine learning under various periods during COVID-19 pandemic [16, 22, 31, 32, 42]. Using machine learning, we have provided an interpolation of what would have happened if the vaccination campaign started earlier during the Olympic Games (Fig. 10(a)). From the computational example, the number of DPC would be much smaller than actual values if the vaccination had started 1 month earlier. Several papers discussed the incidence of COVID-19 new cases during and after the Olympic games [49, 50]. However, our point here is that the number of DPC may be saturated earlier if the vaccination campaign has started earlier. This simulation provides interesting perspective for vaccination planning for such a large event.

The primary limitations of this study are as follows:

1. Due to the limited number of people who were infected in Japan prior to the sixth wave, the population with asymptomatic infection is a major source of uncertainty (see Fig. 3(b)). Even when considering a factor of four to eight for the ratio of asymptomatic infected individuals to the DPC, the population-level immunity in the sixth and seventh waves have varied by 8% and 12%, respectively.
2. The decay of infection-acquired immunity is assumed to be identical to that of vaccination. However, it is well-established that the former is longer than the latter. This simplification does not lead to significant differences because the number of infected people in Japan was limited until the wave we focused on. Similarly, the consideration of hybrid immunity [51, 52] defined as immunity acquired through complete vaccination and infection, was marginal. Nonetheless, the impact of the above-mentioned limitations on the forecasting or the transmission was demonstrated to be marginal through machine learning replication.

6 Conclusions

This chapter numerically discussed the population-level immunity required for transient suppression across the fifth, sixth, and seventh waves of COVID-19 across three major metropolitan prefectures in Japan. A feature of COVID-19 infection in Japan was that only 11% of the population was infected before the seventh

wave. Therefore, the uncertainty factor associated with immunity (such as hybrid immunity and reinfection) was relatively small. The population-level immunity required for transient suppression was consistent across prefectures. The effect of the vaccination rate on positive cases per population was mild for the fifth and sixth waves, when the effect of the restriction was dominant. The correlation between the vaccination ratio and positive cases was more statistically significant for the seventh wave. Meanwhile, viral infectivity and population activity that increased with subsequent waves, characterize the immunity required for suppression at the population level. Our discussion suggested that if viral infectivity is known, then the number of people who can become infected until transient suppression can be roughly estimated. This hypothesis can be confirmed using machine learning. The combination of heuristic derivation and machine learning would be beneficial for policy formulation and the allocation of medical resources, particularly for vaccination strategies.

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